



# **The Global Diabetes Compact - progress in supporting its workstreams**

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**Technical report**



**World Health  
Organization**

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The Global Diabetes Compact - progress in supporting its workstreams: technical report

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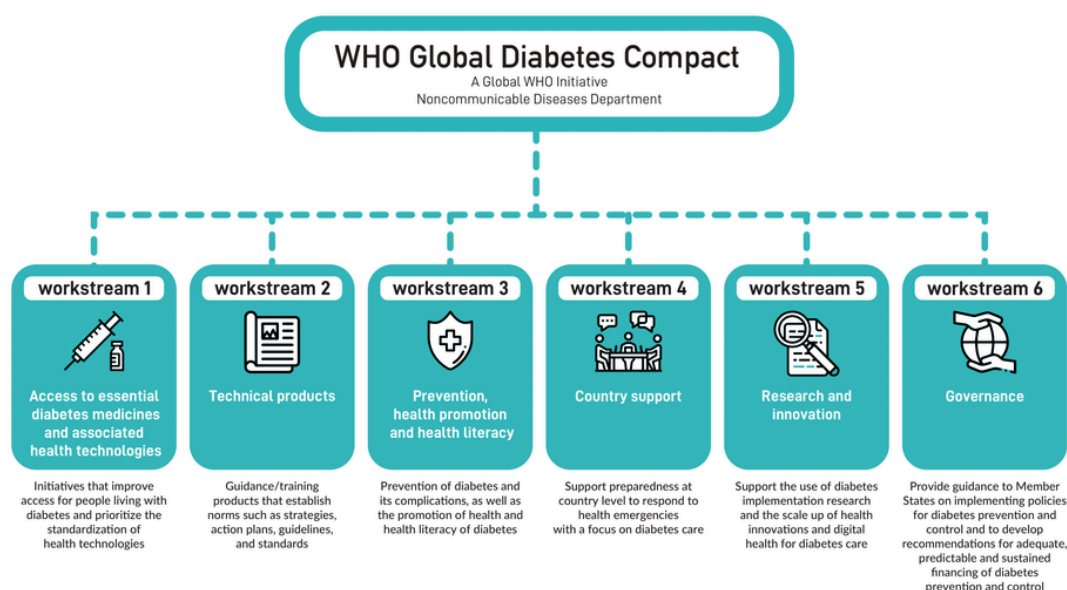
Design and layout by Li Ling Chen

# Introduction

The World Health Organization (WHO) prioritizes five noncommunicable diseases (NCDs) - cardiovascular disease, cancer, mental health, chronic respiratory diseases and diabetes, which share common risk factors and are the leading causes of disease and death globally. NCDs are also included in the Sustainable Development Goals (SDG). It is widely recognized that diabetes is one of the major public health challenges in the 21st century. Very few countries have so far been able to halt the rise in diabetes and the rise has been steepest in low- and middle-income countries (LMICs). Diabetes is among the top 10 leading causes of death. More importantly, it is the only major noncommunicable disease showing an increase in premature deaths between the year 2000 and 2019.

In 2021 WHO launched the WHO Global Diabetes Compact (GDC), an initiative that aims to reduce the risk of diabetes and ensure that all people who are diagnosed with diabetes have access to equitable, comprehensive, affordable and quality treatment and care. To achieve the objective of the GDC, it works through six workstreams; 1) access to essential diabetes medicine and associated health technologies; 2) technical products; 3) prevention, health promotion and health literacy; 4) country support; 5) research and innovation; 6) governance (Figure 1):

**Figure 1.** Workstreams of the WHO Global Diabetes Compact



The GDC is focused on LMICs and is a response to the rising burden and global commitments to reduce premature mortality due to NCDs by 25% by the year 2025 and 33% by the year 2030. The GDC convenes all stakeholders in the diabetes field around a common agenda.

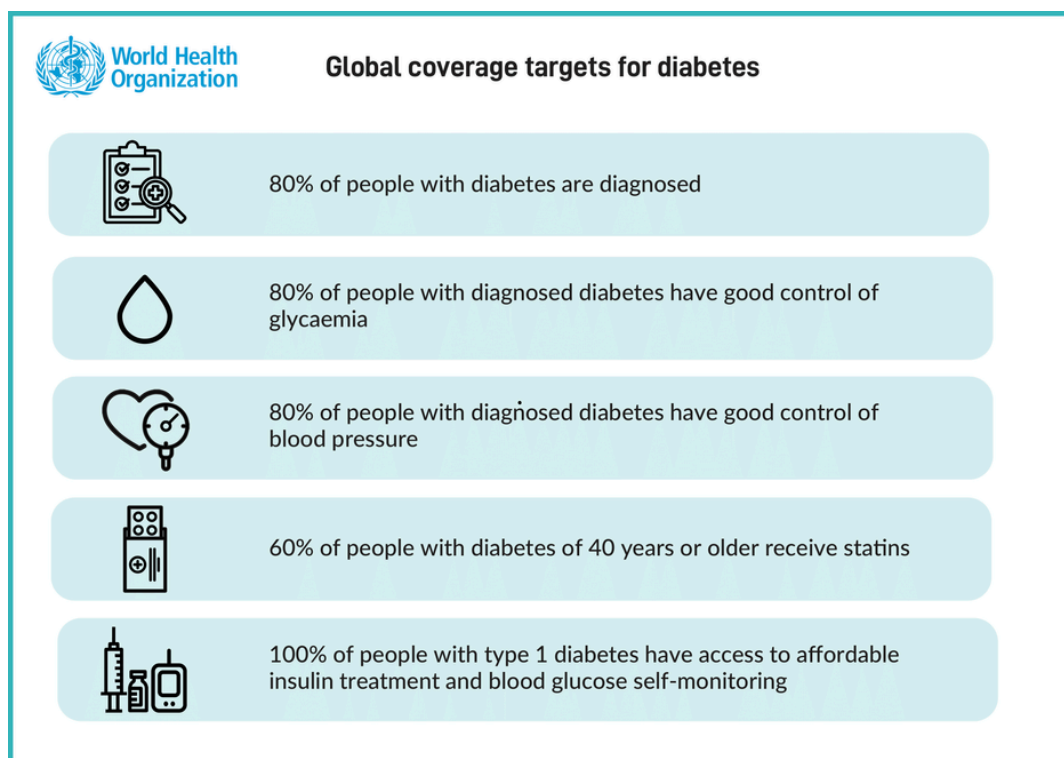
The public health problem of diabetes was recognized with the adoption of Resolution 74.4 by the World Health Assembly (WHA) in 2021. The resolution calls for strengthening prevention and control of diabetes through several mechanisms.

Although globally much is known about diabetes, there are still many gaps in scientific knowledge. Additionally, there is a gap between existing knowledge and its implementation in policy and practice globally, and this is much more prominent in LMICs. The need for more information to guide policy and practice on topics particularly relevant to LMICs was discussed at a series of expert and stakeholder meetings held in 2020.

### Several technical products have been produced within the scope of the GDC:

- a set of five diabetes targets to be achieved by the year 2030 has been formulated and endorsed by the WHA in 2022 (Figure 2)
- a WHO technical brief on the thermostability of human insulin
- a knowledge summary on tobacco and diabetes
- a Cochrane systematic review on diabetes as a risk factor for tuberculosis disease
- updated global diabetes estimates.

**Figure 2.** Global coverage targets for diabetes to be achieved by 2030



### A number of technical products are in the pipeline:

- management of type 1 diabetes in children and adolescents
- insulin use in adults with type 1 diabetes
- updated pharmacological management of type 2 diabetes
- management of dyslipidaemia
- management of hyperglycaemia in pregnancy
- a curriculum for therapeutic patient education
- a global monitoring framework for diabetes
- a global diabetes monitoring framework of diabetes indicators for secondary and tertiary care
- an operational handbook to support countries implement current WHO recommendations and collaborative activities on tuberculosis and diabetes
- a technical brief on oral diseases in diabetes.

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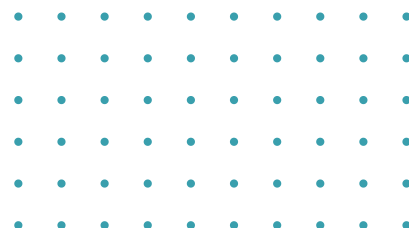
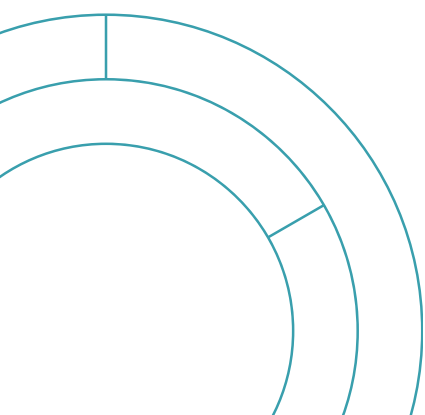
## What is new in the Global Diabetes Compact?

The WHO Global Diabetes Compact has introduced several stakeholder engagement mechanisms. One initiative was a WHO-coordinated process of globally gathering research ideas and evaluating their importance in the context of LMICs in the next 5 years. The approach has been developed by the Child Health and Nutrition Research Initiative (CHNRI) and has been used in a number of similar exercises. It allows systematic listing and transparent attribution of importance to a multitude of research options. A number of research needs were identified at the global stakeholder meetings in 2020. Furthermore, 300 stakeholders in 88 countries were invited through an online survey to submit up to 10 research ideas they thought were important and could be addressed by the year 2030 to impact the diabetes burden in LMICs.

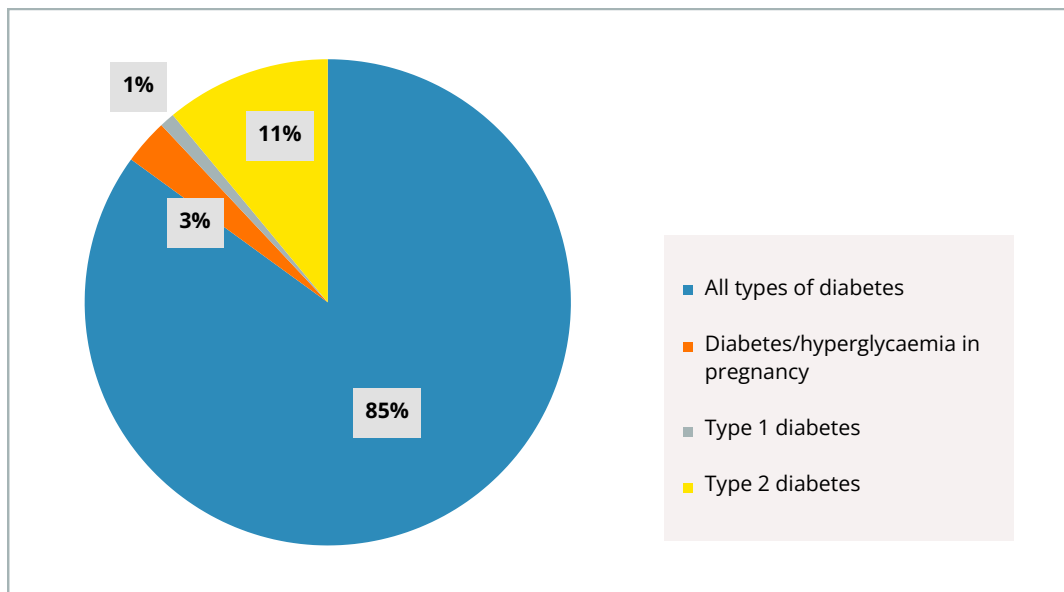
The invited stakeholders included diabetes and public health experts, policy makers, people with diabetes, national diabetes programme leaders, officials of research funding bodies, professional and nongovernmental organizations. They were identified through the GDC Forum membership, WHO Regional Advisers for Noncommunicable Diseases, the International Diabetes Federation (IDF), WHO Collaborating Centres, major international humanitarian agencies, international professional associations, the WHO Technical Advisory Group on Diabetes (TAG-D) and search for leading and/or senior authors of articles published in three major general and four diabetes journals during 2020 and 2021.

In addition, research needs were also extracted from WHO guidelines on diabetes diagnosis, classification and management and identified through a review of the “Implications for research” section of Cochrane Systematic Reviews published on diabetes in the period 2012-2022.

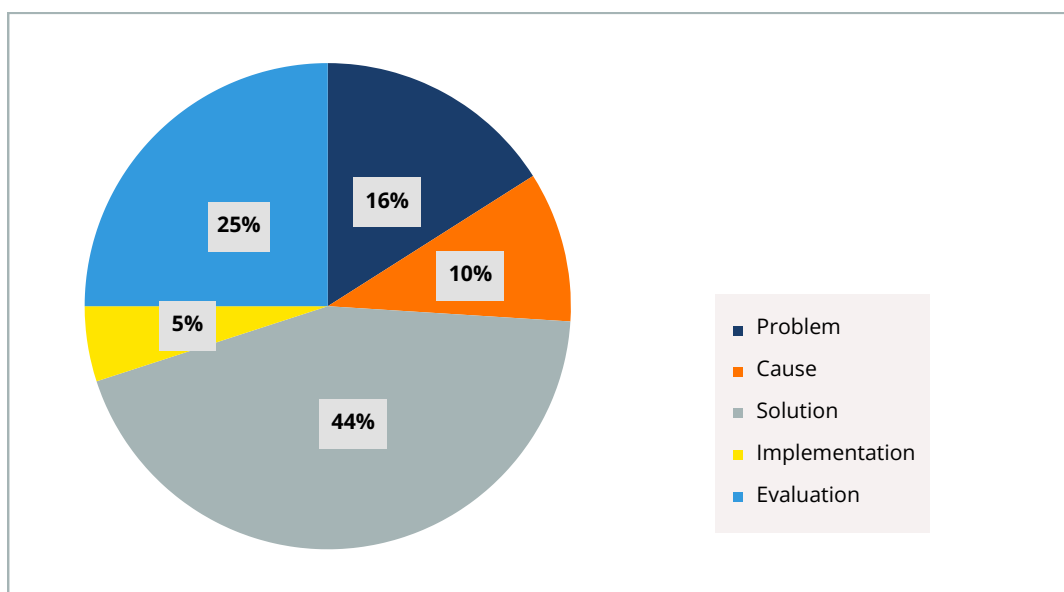
Ninety-four individuals from 45 countries submitted over 500 research ideas. The contributors include a mix of people with different backgrounds and expertise, representing all geographical country distribution and more than one-half coming from LMICs. Additionally, one hundred and fifty-five questions were derived from Cochrane Systematic Reviews and WHO diabetes guidelines. The WHO coordinating group and two diabetes experts from the WHO Collaborating Centre for Physical Activity, Nutrition and Obesity, Boden Institute of Obesity, Nutrition and Exercise and the Prevention Research Collaboration, at the University of Sydney, Australia, curated the questions received through the survey as well as the additional questions from WHO guidelines and Cochrane reviews were curated by removing duplicates, out of scope questions and reconciling different levels of details. The curation of all the received questions resulted in 65 questions. Most of the questions were on all types of diabetes (Figure 3) and on finding solutions for problems (Figure 4).



**Figure 3.** Distribution of 65 research questions by type of diabetes



**Figure 4.** Distribution of 65 diabetes questions by research type <sup>a</sup>



<sup>a</sup> A number of questions were a combination of research types (**Categories of research types - Problem:** measure the size of the health problem through epidemiology, estimating the burden of disease and other forms of data collection; **Cause:** understand the causal agents, risk factors and determinants of the health issue. May include the role of socioeconomic factors, environment, diet and the interaction of multiple factors. **Solution:** develop new interventions. Includes therapeutics, devices and procedures but also includes policy interventions, public health campaigns, etc. **Evaluation:** research to monitor and evaluate the effectiveness or health impact of an intervention or programme)

### Identifying the most important questions

Two hundred seventy-four experts (Table 1), identified by their publication record, were invited to score the curated research questions, using the criteria defined by the WHO Technical Advisory Group for Diabetes (TAG-D) and assumed to be of equal importance (Table 2).

**Table 1.** Distribution of scorers by regions of residence

| <b>Region of residence</b>                   | <b>%</b> |
|----------------------------------------------|----------|
| Western Europe                               | 25.0     |
| South-East Asia and Oceania                  | 20.2     |
| Sub-Saharan Africa                           | 18.3     |
| North America                                | 11.5     |
| Latin America and the Caribbean              | 6.7      |
| North Africa and the Middle East             | 6.7      |
| High-income Asia Pacific and Australasia     | 6.7      |
| Central and Eastern Europe, and Central Asia | 4.8      |

**Table 2.** Criteria for ranking received inputs (defined by TAG-D, 2022)<sup>a</sup>

|                      |                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Significance</b>  | Is it important? (Is the research question an important question that needs answering?)                                                                                                                 |
| <b>Answerability</b> | Can an ethical study be designed to answer this question? (Based on existing methodology and the size of the knowledge gap, can a study be designed to answer the research question in an ethical way?) |
| <b>Feasibility</b>   | Can the study be done? (Is it possible to conduct and finish the research, taking into account the complexity of the research and the infrastructure required?)                                         |
| <b>Impact</b>        | Potential impact on diabetes burden (Does the research have the potential to meaningfully reduce the burden of people with diabetes, care givers and society as a whole?)                               |
| <b>Equity</b>        | Would everyone benefit? (Does the proposed research have the potential to improve equity in disease burden for the underprivileged as well as other segments of society?)                               |

<sup>a</sup>For each research question, participants were asked to select either 'Yes (likely to meet criterion)', 'No (unlikely to meet criterion)', 'I am sufficiently informed, but I am not sure whether it meets the criterion', or 'I am insufficiently informed to make a judgement'.

### Wider stakeholder input

The perspective of non-expert stakeholders, predominantly people with diabetes, but also their carers and family members, policymakers and research funders, was captured in an online global survey disseminated by the International Diabetes Federation. The participants were asked to allocate a value to each of the five scoring criteria, thus providing their opinion on the relative importance of each of the five criteria selected by the TAG-D as being equally important. The mean scores obtained from experts were adjusted by multiplying them with the resulting weights and an overall score for each question was calculated.



Eighty-nine non-expert stakeholders from 35 countries provided their opinion on the importance of each criterion. One-half of the respondents were from LMICs and two-thirds were people with diabetes or their family members/carers. Impact on the diabetes burden and impact on equity were considered more important than the other prioritization criteria. Mean scores obtained from experts for each criterion were adjusted by multiplying them with the weights derived from the wider stakeholders survey.

### *Important diabetes inquiries*

Table 3 shows the 10 questions that achieved the highest scores. The majority of the questions were about finding solutions to known problems and on the implementation of known evidence-based interventions. There was generally very good agreement among experts in their assessments of the criteria (Table 2).

**Table 3.** Top 10 most important questions to facilitate the achievement of the Global Diabetes Compact objectives

| Rank | Research question                                                                                                                                                                                  | Weighted RPS <sup>a</sup> | AEA <sup>b</sup> (%) |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|
| 1    | Defining the best approach to diagnose and manage hyperglycaemia in pregnancy (including GDM) in resource-poor settings                                                                            | 97.59                     | 94.98                |
| 2    | Determining barriers to access to quality, affordable diabetes medicines and related delivery devices and finding solutions to remove barriers                                                     | 97.15                     | 95.81                |
| 3    | Studying barriers to implementation of established and new diabetes interventions and ways to overcome them                                                                                        | 95.84                     | 92.75                |
| 4    | Using community health mechanisms to improve diabetes prevention, diagnosis and management                                                                                                         | 94.93                     | 91.75                |
| 5    | Developing methods for delivering health education to people with diabetes in LMIC and evaluating their impact on patient outcomes                                                                 | 94.65                     | 91.32                |
| 6    | Identifying cost-effective models of diabetes care in low-resource and humanitarian settings                                                                                                       | 94.64                     | 91.76                |
| 7    | Identifying health system challenges for diabetes in LMIC and methods to address them to improve diabetes management                                                                               | 93.70                     | 92.44                |
| 8    | Developing and applying methods/strategies to measure and monitor the health and economic burden of type 1 and type 2 diabetes in LMIC (trends in incidence, prevalence, complications, mortality) | 93.25                     | 90.85                |
| 9    | Developing and evaluating approaches to improve the early detection and prevention of cardiovascular disease and other diabetes complications                                                      | 93.12                     | 88.41                |
| 10   | Investigating ways of improving diabetes prevention and management in primary health care                                                                                                          | 93.01                     | 89.94                |

<sup>a</sup>RPS – research priority score; <sup>b</sup>AEA – average agreement among experts – maximum agreement is 100%.

The 65 consolidated questions can also be grouped and ranked within diabetes themes such as prevention, management, economics, etc. Table 4 shows the questions that are not in the top 10 important questions but are ranked high within their diabetes thematic group.

**Table 4.** Questions ranked highest in each diabetes thematic group but which are not in the top 10 most important questions

| Overarching diabetes theme   | Overall rank | Specific topic                                                                                                                                                                                        |
|------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complications                | 22           | Improving management of diabetic kidney disease                                                                                                                                                       |
| Diagnosis and classification | 47           | Understanding the different presentations of diabetes in LMIC, their underlying pathophysiology, molecular phenotypes, therapeutic response and how they compare between populations/countries        |
| Health systems/services      | 16           | Developing approaches to increase the number and build capacity of health professionals in diabetes prevention, diagnosis and care                                                                    |
| Access to medicines          | 28           | Measuring the impact of inclusion of analogue insulins on the WHO Model Essential Medicines List on analogue insulins price, availability and accessibility                                           |
| Prevention                   | 19           | Testing whether lifestyle interventions reduce diabetes risk in those identified by risk markers other than impaired glucose tolerance (IGT)                                                          |
| Management                   | 12           | Developing and evaluating technology (artificial intelligence, mobiles apps, social media, telehealth) to improve diabetes prevention, diagnosis and management in LMIC                               |
| Monitoring                   | 26           | Establishing indicators for quality of diabetes care in LMIC and methods for their monitoring and evaluation                                                                                          |
| Risk factors                 | 13           | Assessing social determinants of diabetes in LMIC                                                                                                                                                     |
| Economics                    | 25           | Evaluating the cost-effectiveness of opportunistic screening for/early detection of diabetes in LMIC in reducing the proportion of undiagnosed diabetes and assessing its impact on diabetes outcomes |
| Stakeholders                 | 37           | Investigating what people with diabetes prioritize in access to care                                                                                                                                  |

### *Translating findings into policy*

Barriers to translating diabetes research results into policy and ideas for overcoming these barriers was explored by the University of Geneva through a series of interviews with key informants from all over the world. These included policy makers, research funders, managers of research institutions and diabetes advocates. The results of these interviews were presented and discussed at a workshop organized in 2023, with the aim to get insight from various stakeholders on how to attract the interest of different audiences and policy makers in important unresolved diabetes scientific issues.

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Five key themes were identified in the interviews:

1. There is a lack of understanding of diabetes by policymakers outside of the health area. The interviewees also highlighted that for policymakers, diabetes is a complex issue requiring multiple areas to be addressed in parallel.
2. There are different ways of making diabetes a priority.
3. How the “burden” is defined is important.
4. Countries and/or global research funding bodies use a variety of criteria and methods to determine the importance of particular research topics.
5. There is a lack of opportunities for researchers and policymakers to interact. These two stakeholder groups operate in silos and there is a communication barrier between researchers and policymakers.

## Comment

This interaction with global expert and non-expert diabetes stakeholders identified a diversity of unresolved scientific questions of relevance to LMICs. Several topics feature in the top 10 important questions. These include issues around diagnosis, access to medicines, implementation of proven interventions, use of community health mechanisms, education for people with diabetes, delivery of care, economic burden, prevention of cardiovascular disease and other complications, and role of primary health care. Many of the top 10 questions focus on health systems and the delivery of care and also have a possible impact on both type 1 and type 2 diabetes. The qualitative component of this work identified a variety of challenges implementers are likely to face in assisting countries in the choice of research topics to address in their populations.

Insight from the qualitative interviews echo both the multiple facets of diabetes that need to be researched and that these are complex issues. The main barrier mentioned in the interviews and discussed during the workshop was the communication of research results in understandable and useable formats for policy makers. Different knowledge brokers could play a role in bridging this gap. In addition, opportunities for interactions between different stakeholders was seen as essential to further the integration of science into policy and break existing silos.

## Limitations

The main concern in this type of process is the uncertainty over how much the potential selection bias inherent in the process of soliciting inputs and how the self-selection of expert scorers and non-expert stakeholders affects the assessment of the underlying collective opinion of these groups. The numbers of scorers and non-expert stakeholders are generally considered to be sufficient to reliably reflect the opinion of their similarly-minded groups. But it is possible that other groups could have arrived at a different list of top important questions if the selection mechanisms are different. It is reassuring that the non-responders to the invitation to score were significantly more likely to be from high-income and less likely to be from low-income countries than the respondents. Given the context of the Global Diabetes Compact, this difference might even have contributed to a more accurate recognition of topics relevant to LMICs. In any case, it is doubtful that any of the other methods, which have previously been used for identifying the most impactful research topics in the field of NCDs, could have overcome these difficulties.

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## Conclusion

This report provides an update of the technical work undertaken within the second workstream of GDC. It fulfills the request from WHO Member States included in the WHA Resolution on “Reducing the burden of noncommunicable diseases through strengthening prevention and control of diabetes”. The proposed list of important topics offers researchers and funding bodies a choice of topics that the collective opinion of diabetes stakeholders identifies as having potential to meaningfully impact the burden of diabetes in LMICs and many are also pertinent to high-income settings. This report also identifies some areas to address in order to facilitate the translation of research results into policy and practice.

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